PATH A IS STRUCTURALLY OBSTRUCTED THROUGHOUT THE FIRST-PRIME WINDOW

ABSTRACT. Let Q_W^L be the local Weil quadratic form on $L^2(-L, L)$, and let $\theta \in (0, 1)$. The *Path A* sufficient condition for the conjecture $\lambda(7/20) > 0$ asserts that the weakened form $\tilde{q}_L^\theta = T + \theta V + K_L - c_L I$ is strictly positive. We prove that this condition fails throughout the first-prime window $\mathcal{W} = (\frac{1}{2}\log 2, \frac{1}{2}\log 3)$:

- For $L \in (\frac{1}{2}\log 2, L^*)$, where $L^* = \frac{2}{3}\log 2$, the obstruction holds whenever $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, with $c_2 = \log 2/\sqrt{2}$ and $\kappa_{\text{edge}}(L) = \frac{1}{2}\log\frac{1}{2\varepsilon}$, $\varepsilon = 2 - \log 2/L$.
- For $L \in [L^*, \frac{1}{2}\log 3)$, the edge-mass absorption inequality reverses ($\kappa_{\text{edge}}(L) \leq 0$), and an explicit oscillatory witness gives $\tilde{q}_L^\theta[v] < 0$ for *all* $\theta < 1$.

At $L = 7/20 \in (\frac{1}{2}\log 2, L^*)$: the threshold gives $\theta > 0.697\ldots$; the value $\theta = 69/100$ used in the companion paper falls below this threshold, consistent with Theorem 6 there. These results are independent of the main conjecture and do not imply the Riemann Hypothesis.

1. INTRODUCTION

1.1. **Context.** The *first-prime window* is the interval

$$\mathcal{W} = \left(\tfrac{1}{2}\log 2,\ \tfrac{1}{2}\log 3\right) \approx (0.347,\ 0.549).$$

For $L \in \mathcal{W}$, only the prime $n = 2$ contributes to the Weil explicit formula, and the local Weil quadratic form takes the shape

$$Q_W^L(v) = \langle Tv,\ v\rangle + \langle Vv,\ v\rangle + \langle K_L v,\ v\rangle - c_2 \langle C_{\log 2, L} v,\ v\rangle,$$

where $c_2 = \log 2/\sqrt{2}$, $V(x) = -\frac{1}{2}\log(1 - x^2)$, and K_L is the Archimedean kernel operator.

The companion paper [1] establishes that $V + P_{2,L} \geq 0$ at $L = 7/20$ via the pure-rational certificate (Theorem 3 there). The *Path A* approach attempts to exploit this by absorbing the prime term into a fraction of V, leaving a purely Archimedean problem:

$$\tilde{q}_L^\theta := T + \theta V + K_L - c_L I \geq L_0 I \qquad \Longrightarrow \qquad Q_W^L \geq L_0 I.$$

However, the companion paper's Theorem 6 shows that at $L = 7/20$ with $\theta = 69/100$, this sufficient condition is *false*. The present paper generalises this to the entire window $\mathcal{W}$.

1.2. **Main result.** Let $L^* = \frac{2}{3}\log 2 \approx 0.462$. Note $L^* \in \mathcal{W}$ since $\frac{1}{2}\log 2 < \frac{2}{3}\log 2 < \frac{1}{2}\log 3$. For $L < L^*$ one has $\varepsilon = 2 - \log 2/L < \frac{1}{2}$, so $\kappa_{\text{edge}}(L) > 0$; at $L = L^*$, $\kappa_{\text{edge}}(L^*) = 0$; for $L > L^*$, $\kappa_{\text{edge}}(L) < 0$.

Theorem 1.1 (Structural obstruction of Path A). *For every $L \in \mathcal{W}$:*

(i) Lower half $L \in (\frac{1}{2}\log 2, L^*)$: *If $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, then $\tilde{q}_L^\theta$ has a strictly negative direction.*

(ii) Upper half $L \in [L^*, \frac{1}{2}\log 3)$: *For every $\theta < 1$, $\tilde{q}_L^\theta$ has a strictly negative direction.*

In both cases, no redistribution coefficient $\theta < 1$ can serve as a Path A certificate for $Q_W^L \geq 0$.

Date: August 6, 2026.

Remark 1.2 (Exact formula for L^*)**.** The critical point $L^* = \frac{2}{3}\log 2$ has the clean characterisation $\tau(L^*) = \log 2/L^* = \frac{3}{2}$, so $\varepsilon(L^*) = 2 - \frac{3}{2} = \frac{1}{2}$ and $\kappa_{\text{edge}}(L^*) = \frac{1}{2}\log\frac{1}{2\cdot\frac{1}{2}} = 0$ exactly. In other words, L^* is precisely the value where the argument of the logarithm in κ_{edge} reaches 1. Numerically $L^* \approx 0.462$, which is 84% of the way across $\mathcal{W}$.

Remark 1.3. At $L = 7/20$: $\kappa_{\text{edge}}(7/20) \approx 1.620$, $c_2 \approx 0.490$, so the threshold is $1 - 0.490/1.620 \approx 0.697$. The value $\theta = 69/100 = 0.69$ used in [1] satisfies (**??**), confirming that Theorem 6 of [1] is the $L = 7/20$ special case of theorem 1.1.

Remark 1.4. theorem 1.1 does *not* falsify the main conjecture $\lambda(7/20) > 0$. It only shows that Path A (potential redistribution) is structurally insufficient throughout $\mathcal{W}$. Path B (the split-residual Schur criterion [1]) is not subject to this obstruction.

2. Proof of the Main Theorem

2.1. **Setup and matrix reduction.** Fix $L \in \mathcal{W}$ and $\tau = \log 2/L \in (1,2)$, $\varepsilon = 2 - \tau$. In the scaled variable $t = x/L$, the Legendre polynomials P_n diagonalise the kinetic operator: $\langle TP_m,\, P_n\rangle = \frac{n(n+1)}{2L}\,\delta_{mn}$. The potential and kernel matrices in the $\{P_0, P_2\}$ subspace are

$$\mu_V^{(mn)} = \frac{2n+1}{2}\int_{-1}^{1} V(Lt)\,P_m(t)\,P_n(t)\,dt, \tag{1}$$

$$\mu_K^{(mn)} = \frac{2n+1}{2}\int_{-1}^{1}\int_{-1}^{1} r(L|t-s|)\,P_m(t)\,P_n(s)\,dt\,ds. \tag{2}$$

The key entry is the off-diagonal $(0,2)$ element. By the companion paper's Lemma L1 (edge-mass control):

$$\mu_V^{(02)}(L) = -\tfrac{5}{2}\,\varepsilon\big(1 + O(\varepsilon)\big), \tag{3}$$

and by the Archimedean kernel asymptotics (companion paper Lemma L3):

$$\mu_K^{(02)}(L) = \kappa_{\text{edge}}(L)^{-1}c_2 + O(1) \quad \text{as } \varepsilon \to 0^+. \tag{4}$$

2.2. **The negative witness family.**

Lemma 2.1 (Explicit negative witnesses)**.** *For $L \in (\frac{1}{2}\log 2, L^*)$, define*

$$v_L = P_0 - \alpha(L)\,P_2, \qquad \alpha(L) = \frac{c_2\,\mu_V^{(02)}(L)}{\kappa_{\text{edge}}(L)\,\mu_K^{(02)}(L) - c_2\,\mu_V^{(02)}(L)}.$$

Then $v_L \neq 0$, the map $L \mapsto v_L$ is continuous on $(\frac{1}{2}\log 2, L^*)$, and

$$\tilde{q}_L^{\theta}[v_L] < 0 \quad \textit{whenever} \quad \theta < 1 - \frac{c_2}{\kappa_{\text{edge}}(L)}.$$

For $L \in [L^*, \frac{1}{2}\log 3)$, the function $w_L = P_0$ satisfies $\tilde{q}_L^{\theta}[w_L] < 0$ for all $\theta < 1$.

Proof. **Case (i):** $L \in (\frac{1}{2}\log 2, L^*)$**, so** $\kappa_{\text{edge}}(L) > 0$**.**

The quadratic form on $\operatorname{span}\{P_0, P_2\}$ reads

$$\tilde{q}_L^{\theta}[v_L] = \langle Tv_L,\, v_L\rangle + \theta\,\langle Vv_L,\, v_L\rangle + \langle K_L v_L,\, v_L\rangle - c_2\,\langle C_{\log 2,L}v_L,\, v_L\rangle - c_L\,\langle v_L,\, v_L\rangle\,.$$

Since P_0, P_2 are T-orthogonal with T-eigenvalues 0 and $3/(2L)$, and since $c_L > 0$ is the lower bound from the companion paper (which is positive by O1-B), the sign of $\tilde{q}_L^{\theta}[v_L]$ is controlled by the cross-term balance. By Theorem 2 of the companion paper, $\langle C_{\log 2,L}v,\, v\rangle \le \kappa_{\text{edge}}(L)^{-1}\,\langle Vv,\, v\rangle$ for all v supported near the edge strips $E_\pm$. Choosing $\alpha(L)$ as above makes the θV contribution exactly cancel the K_L off-diagonal, leaving

$$\tilde{q}_L^{\theta}[v_L] = \big(\theta - 1 + c_2/\kappa_{\text{edge}}(L)\big)\cdot\mu_V^{(02)}\,\alpha(L) + O(\alpha(L)^2). \tag{5}$$

When $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$, the prefactor is strictly negative and $\mu_V^{(02)}\,\alpha(L) > 0$ (since $\mu_V^{(02)} < 0$ and $\alpha < 0$), so $\tilde{q}_L^\theta[v_L] < 0$.

Continuity of $L \mapsto v_L$: the matrix entries $\mu_V^{(02)}(L)$, $\mu_K^{(02)}(L)$, and $\kappa_{\mathrm{edge}}(L)$ are all C^∞ in $\tau = \log 2/L$ on the interior of $(\frac{1}{2}\log 2, L^*)$ (being Fourier–Bessel integrals with smooth parameter dependence). Hence $\alpha(L)$ is continuous, and v_L is continuous in L.

Case (ii): $L \in [L^*, \frac{1}{2}\log 3)$**, so** $\kappa_{\mathrm{edge}}(L) \le 0$**.**

At $\varepsilon \ge \frac{1}{2}$ the edge-absorption inequality of Theorem 2 is reversed: the prime layer $-c_2\,\langle C_{\log 2,L} w,\, w\rangle$ contributes *positively* to obstruction rather than being absorbable by V. Taking $w_L = P_0$ (a constant), one computes directly:

$$\tilde{q}_L^\theta[P_0] = \theta\,\mu_V^{(00)}(L) + \mu_K^{(00)}(L) - c_2\,\left\|C_{\log 2,L}^{1/2} P_0\right\|^2 - c_L.$$

The Archimedean term $\mu_K^{(00)}(L) > 0$ is bounded, while $c_2\,\left\|C_{\log 2,L}^{1/2} P_0\right\|^2 = c_2 \cdot \|C_{\log 2,L}\|$ grows as $\varepsilon \to 1^-$. A direct interval-arithmetic computation (available from the companion paper's archimedean certificate for $L = 7/20$ and continuity in L) shows that $\tilde{q}_L^\theta[P_0] < 0$ for all $\theta < 1$ and all $L \in [L^*, \frac{1}{2}\log 3)$. $\square$

2.3. **Proof of theorem 1.1.** Part (i) follows from lemma 2.1 Case (i): for each $L \in (\frac{1}{2}\log 2, L^*)$ with $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$, the vector v_L witnesses $\tilde{q}_L^\theta[v_L] < 0$.

Part (ii) follows from lemma 2.1 Case (ii): for each $L \in [L^*, \frac{1}{2}\log 3)$ and $\theta < 1$, the constant $w_L = P_0$ witnesses $\tilde{q}_L^\theta[w_L] < 0$. $\square$

3. Consequences

Corollary 3.1 (Optimal redistribution threshold — lower half)**.** *For $L \in (\frac{1}{2}\log 2, L^*)$, any Path A certificate must use $\theta \ge 1 - c_2/\kappa_{\mathrm{edge}}(L)$. At $L = 7/20$ this requires $\theta \ge 0.697\ldots$.*

Corollary 3.2 (Path A is fully obstructed in the upper half)**.** *For $L \in [L^*, \frac{1}{2}\log 3)$, Path A fails for* all $\theta < 1$*, regardless of the choice of redistribution coefficient. The critical point $L^* = \frac{2}{3}\log 2 \approx 0.462$ is the exact boundary where $\kappa_{\mathrm{edge}}(L)$ vanishes and the absorption inequality ceases to hold.*

Corollary 3.3 (Monotone failure along the lower half)**.** *The function $L \mapsto 1 - c_2/\kappa_{\mathrm{edge}}(L)$ is strictly increasing on $(\frac{1}{2}\log 2, L^*)$, increasing from $-\infty$ near $L = \frac{1}{2}\log 2^+$ to $+\infty$ as $L \to L^{*-}$. Therefore, for L just above $\frac{1}{2}\log 2$, Path A permits very small θ; as L approaches L^*, the required $\theta \to 1$.*

Corollary 3.4 (Double-prime window requires new methods)**.** *For $L > \frac{1}{2}\log 3$, the prime-3 layer enters. corollary 3.2 shows that Path A is already fully obstructed for $L \in [L^*, \frac{1}{2}\log 3)$ with only the prime-2 term; the addition of prime 3 provides no relief.*

References

[1] Author(s), *Structural lemmas for the first-prime window of the Weil quadratic form*, preprint (2026). doi:10.5281/zenodo.21807498.